

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 4.1: Greenhouse Effect and Climate Change — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 4.1: Greenhouse Effect and Climate Change
Driving Question	DRIVING QUESTION: Why are Mount Kenya's glaciers melting and Kenyan farmers' rains becoming unpredictable — and what can the physics of our atmosphere tell us about how to respond?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Is the Greenhouse Effect? — Anchoring Phenomenon Introduction	Students examined paired images of Mount Kenya's Lewis Glacier (1934 vs. present day) showing significant ice retreat, read the testimony of James Muriithi, a maize farmer from Embu County, describing the failure of predictable long rains, and analysed a data strip showing the glacier's area shrinking from approximately 1.35 km ² in 1934 to less than 0.20 km ² in 2023.	Students learned that the greenhouse effect is the process by which greenhouse gases (carbon dioxide, water vapour, methane, nitrous oxide) in Earth's atmosphere absorb longwave infrared radiation emitted by Earth's surface and re-emit it back toward the surface, causing additional warming. They learned that this process — amplified by human activity — is the underlying physics that connects glacier retreat on Mount Kenya to disrupted rainfall patterns experienced by Kenyan farmers.	Students began to explain that both the retreat of Mount Kenya's Lewis Glacier and the unpredictability of the long rains in Embu County can be linked to a common cause: enhanced warming of Earth's surface and lower atmosphere driven by increased concentrations of greenhouse gases. Students constructed an initial diagrammatic model of Earth's energy balance and generated questions for the class Driving Question Board that will be investigated across the remaining six lessons of this unit.
Lesson 2 What Human Activities in Kenya Are Increasing Greenhouse Gases?	We observed NASA and Kenya emissions data showing that agriculture and land-use change (including deforestation) are responsible for more than 75 percent of Kenya total greenhouse gas emissions, and that burning charcoal, cattle farming (releasing methane through digestion), and clearing the Mau Forest are among the largest national sources.	We learned that CO ₂ is released by burning charcoal and by deforestation (when stored carbon in trees is released), that CH ₄ is released by cattle through enteric fermentation, and that N ₂ O is released by agricultural soils. These activities increase the concentration of greenhouse gases in the atmosphere above natural levels.	We explained that increasing the concentration of greenhouse gases in the atmosphere — through Kenyan and global human activities — means more infrared radiation emitted by the Earth surface is absorbed and re-emitted back toward the surface, which intensifies the greenhouse effect. This is why the glacier on Mount Kenya is retreating and why climate patterns affecting Embu farmers are changing: the same physics mechanism identified in Lesson 1 is being amplified by

			human activities happening right here in Kenya every day.
Lesson 3 How Does the Greenhouse Effect Actually Work? — Tracing Energy Through Kenya's Atmosphere	Students observed in the PhET Greenhouse Effect simulation that: (a) removing the atmosphere causes surface temperature to drop dramatically; (b) increasing CO₂ and CH₄ concentrations raises equilibrium surface temperature; (c) CO₂ and CH₄ molecules absorb infrared photons and re-emit them in all directions, while N₂ molecules do not absorb IR photons at all; (d) shortwave (visible) radiation passes through greenhouse gas molecules without being absorbed, but longwave infrared radiation emitted by the surface is absorbed by GHG molecules.	Students learned the physics mechanism of the greenhouse effect: incoming shortwave solar radiation passes through the atmosphere and is absorbed by Earth's surface; the surface re-emits this energy as longwave infrared radiation; greenhouse gas molecules (CO₂, H₂O vapour, CH₄) absorb this IR radiation and re-emit it in all directions, including back toward the surface — this 'back-radiation' raises surface temperature above what it would be with no atmosphere (the natural greenhouse effect); human activities in Kenya (charcoal burning, deforestation of the Mau, cattle ranching) have increased GHG concentrations, enhancing back-radiation beyond natural levels (the enhanced greenhouse effect), which explains the rising temperatures driving Mount Kenya's Lewis Glacier retreat.	Students can now explain: WHY Mount Kenya's Lewis Glacier is melting (enhanced back-radiation from elevated CO₂ and CH₄ concentrations raises atmospheric and surface temperatures above the glacier's equilibrium freezing point, driving ice loss); WHY the Embu County farmer's rainfall is becoming unpredictable (higher surface temperatures increase evaporation rates from Lake Victoria and the Indian Ocean, altering the moisture and energy dynamics of the long-rains system, disrupting the March–May precipitation patterns that Kenyan farmers have relied on for generations). Students can trace a complete causal chain from a specific Kenyan human activity to an observable climate impact on the landscape and farming community.
Lesson 4 How Is Global Warming Changing Kenya's Environment?	We observed that NASA global surface temperature data shows a sharply rising anomaly from 1950 to the present, and that the Lake Victoria water level rose approximately 1.2 metres between 2019 and 2020, flooding homes in Kisumu. We also observed that the Lewis Glacier on Mount Kenya has retreated dramatically since 1934, and that Kericho tea harvests and Embu long rains have become increasingly unpredictable.	We learned that a rise in average global surface temperature — caused by enhanced back-radiation from increased greenhouse gas concentrations — drives cascading effects across multiple Earth systems: it accelerates glacier melt, intensifies the water cycle (increasing evaporation and altering monsoon timing), causes some regions to experience more flooding while others experience more drought, and disrupts the agricultural seasons that Kenyan farmers have relied on for generations.	We explained that the Embu farmer is experiencing disrupted long rains because rising surface temperatures over the Indian Ocean alter evaporation rates and shift the inter-tropical convergence zone, changing when and where moisture-laden winds reach inland Kenya. The same warming mechanism that is melting the Lewis Glacier is intensifying and destabilising the water cycle that governs Kenyan rainfall — so both phenomena share a single physical cause: the enhanced greenhouse effect driven by rising greenhouse gas concentrations.
Lesson 5 How Do Human Activities Amplify the Greenhouse Effect — and What Does This Mean for Mount Kenya and Kenyan Farmers?	Students analysed the Keeling Curve (CO₂ concentration 1958–present) and a global temperature anomaly graph, identifying a strong upward correlation between rising atmospheric CO₂ and rising global average temperatures. Students also used the PhET Greenhouse Effect simulation to observe that increasing greenhouse gas concentrations raises equilibrium surface	Human activities — including burning fossil fuels (petrol, diesel, charcoal), deforestation, and industrial processes — release additional CO₂, CH₄, and N₂O into the atmosphere, increasing greenhouse gas concentrations above natural levels. Higher GHG concentrations mean more infrared radiation emitted by Earth's surface is absorbed and re-emitted back toward the	The observed rise in both atmospheric CO₂ concentration and global average temperature since industrialisation is explained by the physics of infrared absorption: greenhouse gas molecules (CO₂, CH₄, N₂O) absorb outgoing IR radiation that would otherwise escape to space and re-emit it in all directions, including back toward Earth's surface.

	temperature. Students noted that Kenya's Lewis Glacier has lost over 90% of its mass since 1900, and that the Kenya Meteorological Department has recorded a 2–3 week delay in the onset of the long rains since the 1980s.	surface, reducing the amount of energy escaping to space. This creates a net energy gain at Earth's surface, raising average global temperatures. This enhanced greenhouse effect is the physical mechanism behind the retreat of Mount Kenya's Lewis Glacier and the disruption of the long rains experienced by farmers in Embu, Meru, and Nyeri counties.	When human activities increase the concentration of these gases, this absorption is enhanced — the atmospheric 'blanket' thickens — and more energy is retained in the Earth system. The resulting increase in surface temperature drives glacial melt on Mount Kenya by raising ablation rates above accumulation rates, and disrupts regional precipitation patterns by altering atmospheric circulation and the timing of the Inter-Tropical Convergence Zone (ITCZ), which governs Kenya's long and short rain seasons.
Lesson 6 The Ozone Layer: A Separate Shield — and How Its Damage Connects to Mount Kenya's Disappearing Ice	A UV-B irradiance trend graph for Nairobi (1990–2020) showing a peak in the late 1990s and gradual decline post-2000; a world map of the Antarctic ozone hole at maximum extent; and a structured Khan Academy lesson excerpt on ozone chemistry including the catalytic destruction of ozone by CFCs.	The ozone layer is a stratospheric shield (15–35 km altitude) that absorbs incoming UV-B radiation before it reaches Earth's surface; it is chemically distinct from the tropospheric greenhouse gas layer that traps outgoing infrared radiation; CFCs (chlorofluorocarbons) catalytically destroy ozone molecules — one CFC molecule can destroy thousands of ozone molecules; the Montreal Protocol (1987) has successfully reduced CFC production and the ozone layer is gradually recovering; CFCs are also potent greenhouse gases, meaning ozone depletion and the greenhouse effect are distinct but interacting phenomena; and rising UV-B levels in Nairobi during the 1990s are real, measurable evidence of ozone depletion's local impact in Kenya.	The atmosphere performs at least two separate but related protective functions: (1) the ozone layer in the stratosphere absorbs harmful UV-B radiation, and (2) greenhouse gases in the troposphere regulate surface temperature by trapping infrared radiation. Both functions are disrupted by industrial chemicals — CFCs damage ozone directly and also act as greenhouse gases. The primary cause of Mount Kenya's glacier retreat and the disruption of Kenyan farmers' rainfall patterns is the enhanced greenhouse effect (driven by CO₂, CH₄, and other greenhouse gases), not ozone depletion — but ozone depletion adds a second layer of atmospheric vulnerability that affects ecosystems, human health, and indirectly interacts with climate. The Montreal Protocol demonstrates that when the world acts on clear scientific evidence, atmospheric healing is possible — raising the question of whether similar international action on greenhouse gases can produce comparable results.
Lesson 7 Final Model Building, Driving Question Board Closure, and Unit Synthesis: Answering the Mystery of Mount Kenya's Retreating Glaciers	We observed the original anchoring phenomenon images of the Lewis Glacier (1934 versus today) and re-read the Embu farmer testimony. We observed our own cumulative models and those of our peers, noticing where they agreed and where they differed. We observed the Driving Question Board from Lesson 1 and saw how many of	We learned that a single connected causal chain — from human greenhouse gas emissions through infrared trapping to surface warming to glacier retreat and water cycle disruption — explains both the disappearing Lewis Glacier and the unpredictable long rains in Embu. We learned that our models across the class	We explained, in writing and in visual models, the full physics of why Mount Kenya's glaciers are retreating and why Embu farmers can no longer predict the long rains. Our explanation connects solar radiation input, greenhouse gas absorption of outgoing infrared radiation, enhanced warming from human-caused emission

	our original questions could now be answered.	share a strong consensus on the core mechanism while retaining honest differences on details, which reflects how real climate science works. We learned that Kenyan actions — from Olkaria Geothermal to household energy-efficient jikos — represent genuine responses within this causal chain.	increases, glacier mass loss, disruption of the East African water cycle, and Kenya-specific mitigation and adaptation responses — forming a complete, evidence-based scientific account of the anchoring phenomenon first posed in Lesson 1.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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